

Foundation Models & Large Language Models

Complete Study Notes · Generative AI Reference

1. What are Foundation Models?

The term **foundation model** was first coined by a team at Stanford University. They observed that the field of AI was converging toward a new paradigm — instead of building a library of separate, task-specific AI models, the field was moving toward a single foundational model that could drive a wide range of applications.

Large Language Models (LLMs) — such as ChatGPT — are a specific type of foundation model focused on language understanding and generation.

Shift in AI Paradigm

Old Paradigm

Multiple task-specific AI models, each trained on narrow labelled data for one specific task.

New Paradigm

One large foundation model trained broadly, then adapted to drive many use cases and applications.

2. How Foundation Models are Trained

Foundation models are trained in an **unsupervised manner on unstructured data** — typically terabytes of text scraped from the internet, books, articles, and other sources. No manual labelling of data is required at this stage.

The Core Training Task — Next Word Prediction

At its heart, the model is trained to predict the next word in a sentence based on the words it has seen. For example:

Input: "No use crying over spilled ____" Prediction: "milk" The model learns statistical patterns across billions of such examples, building a deep understanding of language structure, facts, reasoning, and context.

This generative capability — predicting and generating the next word — is why foundation models are classified as part of **Generative AI**. They are always generating something new.

3. Adapting Foundation Models to Specific Tasks

Even though foundation models are trained on a generation task (next-word prediction), they can be adapted to perform a wide variety of downstream tasks through two main techniques:

Tuning

Prompt Engineering

Introduce a small amount of **labelled data** to the pre-trained model. The model's parameters are updated to specialise it for a specific task.

Use cases: Classification, Named Entity Recognition (NER), sentiment analysis, question answering, summarisation.

Key benefit: Requires far less labelled data than training from scratch, because the model already has rich pre-trained knowledge.

No labelled data needed. You craft an input (a **prompt**) that guides the model to produce the desired output by completing the text naturally.

Example: 'Is the sentiment of this sentence positive or negative? Sentence: The product is amazing.' → Model outputs: 'positive'

Key benefit: Works in zero-shot or few-shot settings — no fine-tuning required.

Key insight: The same foundation model can be used for classification, entity recognition, summarisation, translation, code generation, and much more — all without retraining from scratch. This transferability is the model's 'superpower'.

4. Advantages and Disadvantages

Advantages

Performance

These models are trained on terabytes of data. When applied to smaller tasks, they drastically outperform models trained on only a few data points.

Productivity Gains

Through prompting or tuning, you need far less labelled data to build a task-specific model than starting from scratch. The model leverages all the unlabelled pre-training data.

Disadvantages

Compute Costs

Very expensive to train due to enormous data requirements. At billions of parameters, even running inference requires multiple GPUs, making them costly compared to traditional approaches.

Trustworthiness

Trained on internet-scraped data that may contain bias, hate speech, or toxic content. Even large annotation teams cannot vet every data point. Dataset details are often unknown for open-source models.

5. Foundation Models Across Domains

Foundation models are not limited to language. The same paradigm is being applied across a wide range of fields:

Language	LLMs built into conversational assistants and enterprise search tools.
Vision	Text-to-image models (e.g. DALL-E 2) generating custom images from text prompts.
Code	Models like Copilot that complete code as it is being authored.
Chemistry	Moleformer — a foundation model for molecule discovery targeting specific therapeutics.

6. LLMs vs Foundation Models — Relationship

Term	Definition	Examples
Foundation Model	Broad AI model trained on large unlabelled data; adaptable to many tasks.	GPT-4, PaLM, BERT, CLIP
Large Language Model (LLM)	A foundation model specifically trained on text data for language tasks.	GPT-4, Claude
Task-Specific Model	A model (often fine-tuned from a foundation model) for a specific task.	Spam classifier, NER tagger

7. Key Takeaways

1	Foundation models are large AI models trained on massive unlabelled data, designed to be adapted to many tasks.
2	LLMs are a subset of foundation models — specifically trained on text for language understanding and generation.
3	The core training task is next-word prediction, which gives models generative capability and broad language understanding.
4	Tuning uses a small amount of labelled data to specialise a foundation model for a specific task.
5	Prompt engineering uses carefully crafted inputs to guide the model without any additional training data.
6	Key advantages: superior performance due to scale, and faster development through less labelled data needed.
7	Key disadvantages: high compute costs for training and inference, and trustworthiness risks from unvetted training data.
8	Foundation models extend beyond language to vision, code, chemistry, climate science, and more.